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13 October 1983
SUPERSEDING
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9 June 1977

MILITARY SPECIFICATION

INDICATOR, MS17264, AIRSPEED AND MACH NUMBER, INTEGRALLY LIGHTED, PITOT-STATIC 80 - 850 KNOTS

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, acceptance, and certain design requirements for the Airspeed and Mach Number, Integrally Lighted, Indicator. The indicator is to have an indicated airspeed range of 80 to 850 knots and Mach number range of 0.4 to 2.5 Mach at altitudes from -1,000 to 80,000 feet.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6610

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SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000/1	Drawings, Engineering and Associated Data
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-L-6085	Lubricating Oil, Aircraft Instrument, Low Volatility
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum Alloys and Aluminum
MIL-C-14806	Coating, Reflection Reducing for Instrument Cover Glasses and Lighting Wedges
MIL-T-18303	Test Procedures; Production, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements for
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-C-26482	Connectors, Electric, Circular, Miniature Quick Disconnect

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*MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-I-81400	Instruments, Aircraft, General Specification for
MIL-G-81704	Glass, Aircraft Instrument, Lighting Wedges and Cover

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedure for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-859	Standard Calibration Table for Aeronautical Pressure Measuring Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-1188	Commercial Packaging of Supplies and Equipment
MIL-STD-1524	Table of Differential Pressure in Relation to Calibrated Airspeed
MIL-STD-2074	Failure Classification for Reliability Testing

DRAWINGS

MS17264	Indicator, Airspeed and Mach Number, Integrally Lighted, Pitot-Static, 80 to 850 Knots
MS27467	Connector, Plug, Electric, Straight, Crimp Type, Bayonet Coupling

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MS28105	Window, Dial-Aircraft Instrument Cover, Glass
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS35206	Screw, Machine-Pan Head, Cross-Recessed, Carbon Steel, Cadmium Plated UNC-2A

2.1.2 Other Government publications. The following other government publications form a part of this specification to the extent specified herein.

Department of Defense Standardization

SD-6	Provisions Governing Qualification

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions shall be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.1.1 Retention of qualification. Retention of qualification consists of periodic verification of compliance of the product with the requirements of this specification. The manufacturer shall provide a letter of certification of compliance to the activity responsible for the Qualified Products List at intervals of not more than two years. The letter shall attest that the qualified indicator is still available from the listed plant and can be produced in the same plant under the same conditions as when qualified. Any changes in parts, materials, design, or processes for manufacture of the indicators shall be reported as required by SD-6 Provisioning Governing Qualification.

3.2 Parts and materials. Selection and use of parts, materials, and processes shall be in accordance with MIL-I-81400. Specifications for items not covered above shall be selected in accordance with MIL-STD-143.

3.3 Materials. Materials for which there are no applicable specifications shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Critical materials. Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

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3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.3 Metals. Metals shall be the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to salt spray or atmospheric conditions to which the indicator may be subjected when in storage or during normal service life.

3.3.3.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other.

3.3.3.2 Magnesium alloy parts. Magnesium alloy parts shall be treated in accordance with MIL-M-3171. When abrasion resistance is a factor, an anodic treatment approved by the procuring activity shall be used.

3.3.3.3 Aluminum alloy parts. Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Aluminum alloys which do not anodize as required shall be coated with a chemical film in accordance with MIL-C-5541.

3.3.3.4 Iron and steel parts. Where practicable, iron and steel parts shall be cadmium, chromium or nickel plated in accordance with QQ-P-416, QQ-C-320, and QQ-N-290, respectively.

3.3.4 Protective Treatment. When materials are used in the construction of the indicator that are subject to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip, or scale during normal service life or are affected by extremes of atmospheric or environmental conditions, shall not be used.

3.3.5 Lubricants. When required, the lubricant used in the indicator shall be in accordance with MIL-L-6085. The amount of lubricant used shall be sufficient for at least 1,000 hours operation.

3.3.6 Fungus-proof materials. Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity.

3.3.7 Corrosive fumes. The materials used in the indicator shall not liberate deleterious fumes.

3.4 Design and construction. The indicator shall consist of a pitot-static operated indicated airspeed mechanism which will drive a rotating disc-pointer with a window to show indicated airspeed on a fixed dial as shown in Figures 1 and 2. The indicator shall also contain a static pressure operated altitude mechanism which will drive a moving scale to indicate Mach number. The gearing between the moving scale and the altitude mechanism shall be such that the airspeed pointer will indicate mach number on the moving scale at any combination of indicated airspeed and altitude within the range of the

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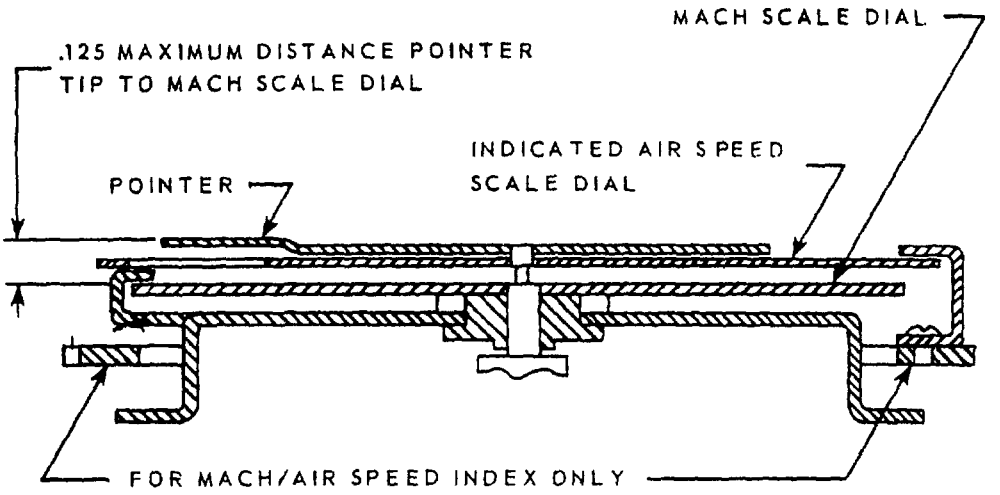
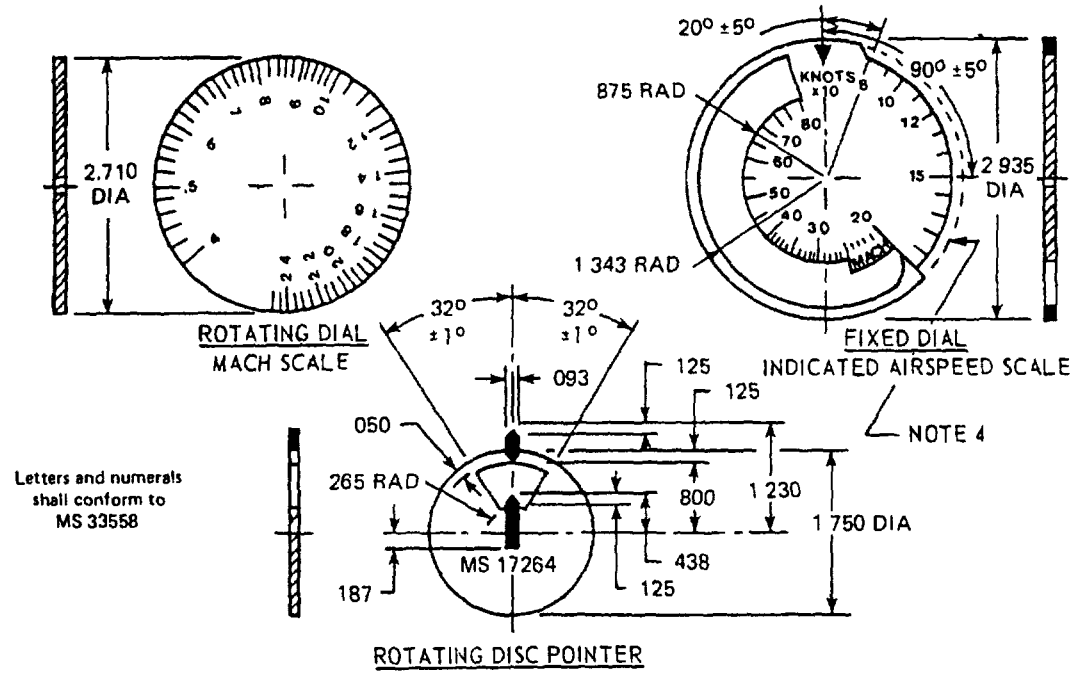


FIGURE 1. Cross section of dial and pointer arrangement with case removed.



1. Numerals shall directly indicate the graduation to which each applies.
2. All graduations are partial radial lines.
3. Dimensions in inches. Unless otherwise specified, tolerances: Decimals ± 0.10, Angles ± 0° 15'.
4. Optional shape when mach/airspeed index not specified.

FIGURE 2. Rotating disc pointer and dial.

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instrument. The design shall be reasonably simple to permit service, maintenance, repair or overhaul and shall comply with the response characteristics and perform as required when subjected to the tests specified in Section 4. Further, the indicator shall be constructed to have a high degree of trouble-free and adjustment-free operation with no parts working loose in service, and shall withstand the normal shocks, vibrations, and such other conditions as are incident to service, shipping, storage, and installation, without functional failure.

3.4.1 Range. The indicator shall be designed to operate as required over a range of indicated airspeed of 80 to 850 knots at altitudes from -1,000 to 80,000 feet at rates of up to 50,000 feet per minute. The mach number range shall be from 0.4 to 2.5 at rates of change in airspeed up to 700 knots per minute.

3.4.2 Maintenance. The design of the indicator shall be such as to accomodate, to the greatest possible extent, the disassembly, reassembly, and service maintenance of the indicator by those tools and items of maintenance equipment which are normally available as commercial standards.

3.4.3 Case. The outline dimensions of the indicator case shall conform to the applicable version of MS17264. The case shall be pressure tight and shall be so designed that the pressure connection is centered and the static connection is off center as shown by MS17264.

3.4.3.1 Body and mounting flange. The body and mounting flange of the case shall be made of lightweight nonmagnetic metal, uniform in texture, and shall have a smooth surface. The body shall be finished in a durable black finish and the mounting flange shall be finished with a durable lusterless black finish Color No. 37038 of FED-STD-595.

3.4.3.2 Pitot and static bosses. The pitot and static bosses shall be in accordance with MS17264 and shall be located as shown thereon. A 150-wire mesh filter screen (either stainless steel or monel) shall be firmly installed in the inserts to prevent dirt particles from entering the case. For storage, shipping, and handling purposes, a vented plug of acceptable material and design shall be provided. The plug shall incorporate a vent consisting of a hole at least 1/16 inch in diameter and shall also incorporate a suitable air filtering material.

3.4.3.2.1 External connections. When external connections to simulate installed conditions are called for, they shall consist of 10 feet of copper, aluminum, or tygon tubing so coiled that the moisture can drain out of the open end. Similar coils are to be connected to the pitot and static pressure connections except that the tubing ID for pitot shall be 1/4 inch and for the static it shall be 3/4 inch.

3.4.3.3 Case Reinforcement. The inside of the case around each boss and the electrical receptacle shall be sufficiently strong to avoid damage to the case which may occur due to excessive torque applied to the connections during installation or removal of the indicator.

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3.4.3.4 Case marking. Case markings shall be made with a lusterless white material of Color 37875 of FED-STD-595. The material shall be sufficiently durable to withstand usage encountered in service. The form of the letters and numerals shall be in accordance with MS33558. The back of the case shall be marked as follows:

a. The letter "P" shall appear both above and below the pitot pressure boss as shown on MS17264.

b. The word "STATIC" shall be printed twice, forming a ring around the static pressure boss as shown in MS17264.

c. The following caution notes shall be legibly marked on the back of the case:

USE ONLY 5 VOLTS

USE STRAIGHT THREAD FITTINGS ONLY

CAUTION: DO NOT BLOW IN TUBES

3.4.4 Application of pressure. The pitot pressure shall be applied to the inside of the diaphragm capsule and the static pressure to the interior of the case.

3.4.4.1 Capillary tubing. A sufficient length of approved capillary tubing having a minimum inside diameter of 0.020 inch, or some other acceptable restriction, shall be incorporated in the indicator case between the pitot connection in the rear of the case and the diaphragm capsule. The effect of this restriction shall be such that the indicator will meet the requirements of the Damping Test specified herein.

3.4.4.2 Standard pressure. The standard airspeed pressure used in calibrating the indicator shall be in accordance with the values specified in MIL-STD-1524. Altitude pressure shall be in accordance with MIL-STD-859.

3.4.5 Weight. The weight of the indicator shall be not greater than 1.75 pounds.

3.5 Lighting, integral. The instrument is to be integrally lighted either Red, White, or Instrument and Panel Lighting White (IPL White) as applicable. The color shall be specified by the procuring activity (see 6.2.1). The lighting circuit shall be wired as specified in MS17264. The case shall not be used for grounding of the lighting circuit. The connector for integral lighting shall be in accordance with MS27467.

3.5.1 Red. When red lighting is specified by the procuring activity it shall be in accordance with MIL-L-25467.

3.5.2 White. When white lighting is specified by the procuring activity it shall be in accordance with MIL-L-27160.

3.5.3 IPL White. When IPL White lighting is specified by the procuring activity it shall comply with the requirements of MIL-C-25050 and MIL-L-25467

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to the extent applicable. Color shall be as required by MIL-C-25050 and brightness shall be in accordance with MIL-L-25467 except that the applied voltage shall be 4.5 ± 0.1 volts dc.

3.5.4 Lamps. The lamps used in the lighting system shall be contained within the case, shall be 5 volts incandescent, and shall consist of the minimum number practicable to produce the required illumination. All lamps shall be in accordance With MIL-L-25467 or MIL-L-27160 whichever is applicable. The lamps shall be removable and shall not be soldered or otherwise cemented to the case. Soldering shall not be required to replace lamps or lamp assemblies.

3.5.5 Electrical connector. The electrical connector shall be in accordance with MS17264 and MIL-C-26482.

3.6 Dial, index, and pointer arrangement. The dials, indices, and pointer shall be arranged as shown in Figure 1.

3.6.1 Dials. The indicated airspeed scale dial and the mach scale dial shall conform to Figure 2 and Table I. The dials shall be of aluminum alloy and shall be fastened securely into the case or upon the frame of the mechanism using screws, or other positive mechanical means, such that they will not loosen or turn when the indicator is vibrated. The use of tape as a means of securing the dials is prohibited. The mechanism used to secure the dials shall in no way interfere with any dial marking. The dials shall not be sensitive to acceleration.

3.6.2 Dial markings. The dials shall be marked as shown in Figure 2, special notice should be taken of note 1 of Figure 2. Unless otherwise specified, all visible portions of the dials shall be lusterless black, Color No. 37038 of FED-STD-595, and all markings on the dials shall be lusterless white, Color No. 37875 of FED-STD-595.

3.6.3 Visibility of dials. The pointer, numerals, at least 1/16 inch of the shortest graduations, and other specified markings on the dials shall be visible from any point within the frustum of a cone whose side makes an angle of 20 degrees with a perpendicular to the dial and whose small diameter is the aperture of the indicator case.

3.6.4 Pointer. The pointer shall conform to Figures 1 and 2 and shall be suitable for integral lighting. It shall be firmly attached to the mechanism to prevent slippage and shall be readily removable during overhaul. The pointer shall not be sensitive to acceleration.

3.6.4.1 Mach number setting index. When specified, a triangular Mach number setting index shall be provided as shown in MS17264. The setting of the Mach number index shall be made by a knob located in the lower left hand corner of the bezel. The index shall be settable over a range corresponding from 0.4 to 2.5 Mach.

3.6.4.2 Airspeed setting index. When specified, a triangular airspeed setting index shall be provided as shown in MS17264. The setting of the index shall be controlled by the knob which controls the Mach number index setting.

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Setting of either this index or the Mach Number index shall not in any way affect the setting of the other. The index shall be settable over a range corresponding to 80 to 190 knots.

3.6.5 Limitation of indicating pointer movement. The pointer movement shall be limited by means of suitable stops in the mechanism in such a way that the pointer will not be permitted to rotate more than 355 degrees of arc clockwise from the zero graduation when the indicator is subjected to over-pressure tests. The design of the stops shall be subjected to the approval of the procuring activity.

3.6.6 Pointer and indices finish. Shaded portions of the rotating disc pointer indices shall be lusterless white, Color No. 37875 of FED-STD-595. The unshaded portion of the pointer shall be lusterless black, Color No. 37038 of FED-STD-595.

3.6.7 Cover glass mounting. The indicator cover glass and any other transparent element between the glass and the dial, such as lighting wedge, shall be in accordance with MIL-G-81704.

3.6.7.1 Cover glass. The quality and dimensions of the cover glass shall be in accordance with MS28105. If the cover glass forms a part of the integral lighting system, then it may deviate from the shape of MS28105 to the extent necessary for that system. All reflecting glass surfaces shall be provided with a reflection reducing coating in accordance with MIL-C-14806.

3.6.7.2 Cover glass mounting. The distance between the front surface of the mounting flange and the outside surface of the cover glass shall not exceed 0.062 inch. The distance between the mach scale dial and the inside surface of the cover glass shall be the minimum practicable and shall not exceed 0.25 inch. The cover glass shall be properly sealed and fogging on the inside surface shall not occur under operational conditions. The cover glass shall be replaceable at overhaul by removing the bezel. -Provisions shall be made for alignment of lighting wedges to facilitate assembly at overhaul.

3.7 Reliability.

3.7.1 Operating life. The equipment shall have a minimum total operating life of 8,000 hours with minimal servicing and replacement of parts.

3.7.2 Reliability in mean time between failure (MTBF) - The indicator shall have a specified MTBF (O_o) of 1,000 hours when tested and accepted in accordance with 4.4.3 and 4.7.

3.8 Nameplate. A nameplate shall be securely attached to the exterior of the case and shall be marked in accordance with MIL-STD-130.

3.8.1 Manufacturer's part number. The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing covers more than one part and lists the part number.

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TABLE I. Pointer and dial markings

Marking	Height or Length ± .010	Width of Line ± .002	Finish
Numerals on indicated airspeed scale dial	.125	--	
Graduations on indicated airspeed scale dial corresponding to:			
10 knot increments 80 to 190 inclusive	.125	.030	
10 knot increments 210 to 390 inclusive	.094	.020	
50 knot increments 450 to 850 inclusive	.125	.025	
100 knot increments 300 to 800 inclusive	.188	.030	
100 knots, 150 knots	.188	.030	
250 knots, 350 knots	.156	.025	
200 knots	.688	.035	
Zero indication - isosceles	.062 base		Luster-
Mach/Airspeed Index triangle	.125 height		less
Numerals on Mach scale dial	.188	--	White 36875
Graduations on Mach scale dial corresponding to:			
.02 increments	.125	.020	FED-STD
.05 increments	.156	.025	-595
.1 increments .4 to 2.5 inclusive	.188	.030	
Dial			
Lettering "KNOTS X 10"	.125	--	
Lettering "MACH"	.094	--	
Shaded portions of rotating disc pointer	--	--	
Unshaded portions of rotating disc pointer	--	--	Luster-
Background of dials and knob	--	--	less Black 37038 FED-STD -595
MS 17264	.125		

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3.9 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DoD-D-1000/1.

3.10 Installation instructions. The contractor shall furnish with each indicator one printed copy of instructions with illustrations and diagrams, if necessary, covering the installation of the indicator. The instructions shall be printed on 8-1/2 by 11 or 11 by 17 inch paper.

3.10.1 Mounting screws. Mounting screws shall be in accordance with MS35206 (see 6.2.1).

3.10.2 Envelope. An envelope furnished by the contractor containing installation instructions shall be packaged with each indicator and shall be marked with the following information:

"IMPORTANT
THIS ENVELOPE CONTAINS INSTRUCTIONS"

When mounting screws are to be provided in addition to the installation instructions, the envelope shall be labeled as follows:

"IMPORTANT
THIS ENVELOPE CONTAINS INSTRUCTIONS AND MOUNTING SCREWS"

3.11 Workmanship. The indicator, including all parts and accessories, shall be constructed and finished to produce an instrument free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of welding and brazing, marking of parts and assemblies, painting, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.11.1 Dimensions. Dimensions and tolerances not specified herein shall be held to the values necessary to ensure that the indicator will conform to all mechanical and electrical requirements.

3.11.2 Fabrication. Machining, drilling, and forming shall be done with the use of accurate templates, jigs, or gages.

3.11.3 Soldering. Soldering shall be performed in accordance with MIL-S-6872.

3.11.4 Screw threads. Screw threads 0.060 inch in diameter or larger, shall be in accordance with MIL-S-7742.

3.11.5 Moving parts. Moving parts shall be properly aligned and shall be operated without interference, looseness, or other irregularities. Bearings shall be as near frictionless as practicable. When required for accurate adjustment, gear assemblies shall be free from detrimental backlash.

3.11.6 Wire stripping. Wire stripping shall be done without nicking or otherwise damaging the wire or insulation.

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3.11.7 Cleaning. The indicator shall be thoroughly cleaned of loose, spattered or excess solder, flux, metal chips, insulation and foreign material at final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The manufacturer shall ensure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 Qualification inspections. The qualification tests shall consist of all the tests of 4.6 and the Reliability Qualification Test Phase, 4.7.1, of the Reliability Test, 4.7.

4.3.1 Qualification samples. For purposes of qualification the manufacturer shall forward three indicators, at his own expense, to the laboratory designated in the letter of authorization (see 6.3). The manufacturer shall also provide three indicators for the Reliability Qualification Test Phase which is to be conducted at his own expense and facility under Government surveillance. All Qualification Test samples shall have been previously subjected only to the Individual Tests.

4.3.1.1 Qualification test sample identification. The qualification test samples shall be plainly identified by durable tags, securely attached, and marked with the following information.

Sample for Qualification Test
(Name of Equipment)
Submitted by (Manufacturer's name, date and part number)
for Qualification Test in accordance
with MIL-I-23671B under
authorization (reference letter authorizing tests.)

4.4 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing all tests except Sampling Plan "B" Tests which will be conducted at a government laboratory designated

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by the procuring activity (see 6.2.1). Quality conformance test, except for Sampling Plan "B", shall be under the supervision of the Government quality control representative. The contractor shall prepare and have available for review test reports showing detailed quantitative results for all tests required by this specification, signed by the authorized Government quality control representative (see 6.2.2). Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Quality conformance tests shall consist of the following tests:

a. Individual Tests (see 4.4.1)
b. Sampling Plan "A" Tests (see 4.4.2.1.1)
c. Sampling Plan "B" Tests (see 4.4.2.2.2)
d. Reliability Tests (see 4.4.3)

4.4.1 Individual tests - Each instrument submitted for acceptance shall be subjected to the Individual Tests. These tests shall determine compliance with the requirements of material, workmanship, and operational performance. As a minimum, each instrument accepted shall have passed the tests of Table II. The Manufacturing run-in test shall be performed prior to the performance of any operational tests.

TABLE II. Individual tests.

Test	Method Paragraph
Examination of Product	4.6.1
Leakage	4.6.2
Mach and Airspeed Pointer Operation	4.6.3
Friction and Scale Error at Room Temperature	4.6.4 -
Low Temperature Operation	4.6.5
High Temperature Operation	4.6.6
Lighting Individuals	4.6.7
Position Error	4.6.8
Magnetic Effect	4.6.9
Manufacturing Run-in	4.7.2

4.4.2 Sampling plan A and sampling plan B tests. The Sampling Plan Tests shall consist of Sampling Plan "A" and Sampling Plan "B Tests. The test samples selected for sampling tests shall first have passed the Individual Tests. The test samples which have been subjected to Sampling Plan "A" Tests shall not be delivered on contract until they have been refurbished and resubmitted and passed all the Individual Tests. Test samples which have been subjected to the Sampling Plan "B" Tests shall not be delivered on contract.

4.4.2.1 Sampling plan "A" sample selection - Sampling Plan "A" samples shall be selected at random and shall consist of one sample from the first 15 indicators produced with an additional sample of one for each additional 15 indicators produced or portion thereof.

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4.4.2.1.1 Sampling plan "A" tests. Each sample selected for Sampling Plan A testing shall be subjected to the tests of Table III.

TABLE III. Sampling plan A tests.

Tests	Paragraph
Damping	4.6.10
Overpressure	4.6.11
Seasoning	4.6.12
Lighting	4.6.13
Vibration Test (Functional)	4.6.16.1

4.4.2.2 Sampling plan "B" tests. Three instruments shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. These samples shall be forwarded at the contractor's expense to the Government laboratory designated by the procuring activity. Each sample shall be plainly identified by a durable tag, securely attached and marked with the following information:

Name of Equipment
Submitted by (Manufacturer's name, date)
for Production Acceptance Sampling Plan "B"
Test of MIL-I-23671 in accordance with
Contract/Order No. _____
Manufacturer's Part No. _____

4.4.2.2.1 Sampling plan "B" approval. No indicators, other than the Sampling Plan "B" units, shall be delivered until the Individuals and Sampling Plan "A" Test results are approved. Approval of Sampling Plan "B" instruments shall be by the procuring activity upon satisfactory completion of the designated tests. Any design, material, or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the sample units to pass any of the tests shall be cause for deliveries of instruments under the contract to cease until proper corrective action is approved and accomplished.

4.4.2.2.2 Scope of sampling plan "B" tests - Each sample selected for Sampling Plan Test shall be subjected to the tests of Table IV. Sampling Plan "A" Tests shall be completed prior to the performance of any tests considered destructive.

4.4.3 Reliability test. The reliability test of 4.7 shall be conducted in accordance with MIL-STD-781 using approved test procedures of 4.8.

4.4.4 Rejection and retest. When one item selected for the Individual, Sampling Plan, or Reliability Tests fails to meet the test requirement, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined and appropriately corrected. After corrections have been made, all necessary tests shall be repeated.

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TABLE IV. Sampling plan B tests.

Test	Method Paragraph
Magnetic Effect	4.6.9
Mach Number Accuracy Test	4.6.4.2.2.1
Low Temperature Exposure	4.6.14
High Temperature Exposure	4.6.15
Vibration Test	4.6.16
Acceleration Sensitivity	4.6.17
Acceleration	4.6.18
Lighting	4.6.19
Humidity	4.6.20
Mounting Lugs	4.6.21
Internal Examination	4.6.22

4.4.4.1 Continuation of individual tests. Individual tests may continue pending the investigation of a Sampling Test or Reliability Test failure, but final acceptance of indicators on hand or later produced shall not be made until it is determined that all the requirements of this specification are met.

4.5 Test conditions. Unless otherwise specified, all tests required by this specification shall be made under the following conditions:

4.5.1 Environmental conditions. Standard ambient conditions.

a.	Temperature	Room ambient, 25°C ± 5°C
b.	Pressure	Normal atmospheric (approx. 29.92 in. hg.)
c.	Humidity	Room ambient, 40 to 90% relative humidity

4.5.2 Test reading vibration levels. Unless otherwise specified, the indicator shall be tested in normal operating position and shall be lightly vibrated before a test reading is taken. Vibration shall be applied in the axis perpendicular to the dial at a maximum frequency of 120 Hertz at a maximum acceleration level of 0.2g.

4.5.3 Attitude. The indicator shall be tested in normal operating position.

4.5.4 Integral lighting. The indicator lighting system shall be energized and shall operate as required during all tests except Humidity, Fungus Resistance and Salt Spray, unless otherwise specified herein.

4.5.5 Test points and tolerances. The indicator test points and tolerances shall be as specified in Tables V, VI, and VII, and as otherwise specified herein.

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TABLE V. Test values.

Tests and Test Points	Tolerance ±
Friction: Indicated airspeed pointer 80 to 200 knots 200 to 850 knots Mach scale	 3.0 knots 5.0 knots 5.0 knots
Position Error: Indicated airspeed pointer 80 to 200 knots 200 to 850 knots Mach scale Pointer Zero Position	 3.0 knots 5.0 knots 5.0 knots 1/16 inch
Vibration (Functional): Test altitude - 35,000 feet Indicated airspeed - 350 knots Maximum vibration - Pointer Maximum variation - Pointer Test altitude - Sea Level Indicated airspeed - 120 knots Maximum vibration - Pointer Maximum variation - Pointer	 5.0 knots 5.0 knots 2.0 knots 2.0 knots
Overpressure: Static connection - 6.0 inches water Pitot connection - 60 inches Hg	2.0 knots
Vibration Failure: (Endurance) Test altitude - 35,000 feet Indicated airspeed - 350 knots	2.0 knots
Damping: Time required for the airspeed pointer to travel from full scale deflection to the 90 knots graduation	$1.0^{+1.0}_{-0.2}$ seconds
Aging	3.0 knots

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TABLE VI. Airspeed scale error test points and tolerances

Airspeed Knots	Equiv. in Hg.	Room Temperature Tolerance ± Knots	High and Low Temp. Tolerance ± Knots
80*	0.3075	2.0	4.0
90*	0.3895	2.0	4.0
100*	0.4814	2.0	4.0
110*	0.5832	2.0	4.0
120*Δ	0.6950	2.0	4.0
130*	0.8168	2.0	4.0
140*	0.9488	2.0	4.0
150	1.0910	2.5	4.5
160	1.2345	2.5	4.5
170	1.4064	2.5	4.5
180*	1.5799	2.5	4.5
190	1.7640	2.5	4.5
200*	1.9589	3.0	5.0
220	2.3816	3.0	5.0
250*Δ	3.1002	3.0	5.0
280	3.9240	3.0	5.0
300*	4.5343	4.0	6.0
320	5.1953	4.0	6.0
350*	6.2859	4.0	6.0
380	7.5015	4.0	6.0
400	8.3850	5.0	7.0
450*	10.8674	5.0	7.0
500*Δ	13.7756	6.0	8.0
550	17.1590	6.0	8.0
600*	21.0749	7.0	9.0
650*	25.5893	7.0	9.0
700*	30.7642	7.0	9.0
750*	36.5662	7.0	9.0
800	42.9378	7.0	9.0
850*	49.8423	7.0	9.0

* see 4.6.4.1.1 (Individual Test Points)
Δ see 4.6.8 (Position Error Test Points)

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TABLE VII. Mach number scale error test points and tolerances

MACH NO	ALTITUDE FEET	ALTITUDE INCHES Hg(Ps)	MACH NO TOLERANCE ±	EQUIVALENT PRESSURE INCHES Hg(Pt) LO	EQUIVALENT PRESSURE INCHES Hg(Pt) NOM	EQUIVALENT PRESSURE INCHES Hg(Pt) HIGH	INDICATED AIRSPEED (REFERENCE)
0.6	-1000	31.018	0.015	39.1063	39.5635	40.0365	403
0.4	0	29.921	0.015	33.1423	33.4083	33.6863	264
0.6			0.015	37.7232	38.1642	38.6205	397
0.8			0.015	44.9404	45.6098	46.2992	529
0.9*			0.015	49.7939	50.6054	51.4402	595
1.0*			0.015	55.6605	56.6384	57.6431	661
1.2			0.015	70.7404	72.0348	73.3504	793
0.7	5000	24.896	0.015	34.0781	34.5332	35.0028	426
0.9			0.015	41.4314	42.1066	42.8012	551
1.1			0.015	52.1457	53.0994	54.0724	676
0.9*	10000	20.577	0.015	34.2438	34.8019	35.3760	507
1.0*			0.015	38.2784	38.9508	39.6418	566
0.9	15000	16.886	0.015	28.1013	28.5593	29.0304	464
1.0*			0.015	31.4122	31.9640	32.5310	520
1.1			0.015	35.3684	36.0153	36.6752	577
0.8	20000	13.750	0.015	20.6521	20.9597	21.2765	373
0.9			0.015	22.8825	23.2554	23.6390	423
1.4			0.015	41.2096	41.9271	42.6532	675
0.9*	30000	8.885	0.015	14.7862	15.0272	15.2751	346
1.4			0.015	26.6289	27.0925	27.5617	564
0.9	35000	7.041	0.015	11.7175	11.9084	12.1049	310
1.0			0.015	13.0980	13.3281	13.5646	350
1.1			0.015	14.7477	15.0174	15.2936	391
0.9*	40000	5.538	0.075	3.6566	9.3664	10.1842	277
1.4			0.075	15.4775	16.8867	18.3842	459
1.2*	60000	2.118	0.10	4.5174	5.0991	5.7474	245
1.8			0.10	8.9461	9.8900	10.8903	386
1.8	80000	0.810	0.25	2.9365	3.7823	4.7787	245

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4.6 Test methods.

4.6.1 Examination of product. Each indicator shall be examined externally to determine conformance with the requirements of Section 3 and applicable drawings and specifications not covered by tests. Each indicator shall be checked for conformance to MIL-L-25467 and 4.4.1 of this specification.

4.6.2 Leakage.

4.6.2.1 Case leakage. With the static and pitot connections joined together, the indicator shall be subjected to a vacuum of 25 in. Hg., and then to a pressure of 10 in. Hg. When the indicator is isolated at these pressures for periods of 60 seconds, the internal case pressure shall not change by more than 0.1 in. Hg.

4.6.2.2 Pitot diaphragm capsule leak. With the static connection open to atmosphere, the pitot diaphragm shall be pressurized to full scale pointer deflection, sealed off, and shall not leak more than 1 knot in a period of 60 seconds.

4.6.3 Mach and airspeed pointer operation. With the pitot port open to atmosphere, gradually apply vacuum to the static port until a differential pressure of 29 in. Hg is obtained, (approximately 700 knots airspeed and 70,000 ft. pointer and dial rotation), then return to atmosphere at a rate of 10,000 feet per minute while observing both the Mach dial and airspeed pointer movement. Any stickiness, hesitation (except a slight hesitation at approximately the point of the disengagement of the retard) or erratic movement of either airspeed or mach mechanisms is cause for rejection. No vibration shall be applied during this test.

4.6.4 Friction and scale error at room temperature.

4.6.4.1 Airspeed scale.

4.6.4.1.1 Airspeed Individual test. The indicator shall be tested for airspeed scale errors at the points of the scale indicated by asterisks in Table VI. The test shall be made by subjecting the indicator to the pressure specified to produce these readings, first with pitot pressures increasing, then with pitot pressures decreasing. At each test point the pitot pressure shall be held constant while two readings are taken, the first before the indicator is vibrated, the second after. The difference in readings is friction and shall be not greater than the value specified in Table V. The second reading is the scale error and shall be not greater than the value specified in Table VI. When desired by the Inspector, the indicator shall be tested for scale airspeed error at any additional point or points on the scale. The errors at these additional test points shall not exceed those specified in Table VI.

4.6.4.1.2 QPL, sampling plan B test. The indicator shall be tested for friction and airspeed scale errors at the points of the scale indicated in Table VI. The Airspeed Scale Error Test shall be made by subjecting the indicator to pressure sufficient to produce the desired indicated airspeed, first with increasing Pressures, then with decreasing pressure. With

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increasing pressure, the pressure shall be applied gradually to the indicator until the pointer is indicating the desired airspeed. With decreasing pressure, the pressure shall be gradually reduced until the pointer is indicating the desired downscale airspeed. At each airspeed setting, both upscale and downscale, the pressure required to produce the airspeed shall be recorded and this pressure shall be within the limits of Table VI. During this test the indicator shall be vibrated continuously except when friction tests are performed. As a minimum, the indicator shall be tested for friction at every other point marked by an asterisk in Table VI. The pressure shall be increased to bring the pointer approximately to the desired reading and then held constant while two readings are taken; the first before and the second after the indicator is vibrated. The difference between any two readings shall be within the limits of Table VI. At the discretion of the test activity, this test may be performed at the asterisked test points only or by the test method specified in 4.6.4.1 when the test schedule requires this test to be repeated.

4.6.4.2 Mach number scale.

4.6.4.2.1 Mach number individual test. The Mach scale shall be tested for friction at each of the test points indicated by an asterisk in Table VII. The test shall be conducted, at each of the specified test points, by establishing the specified altitude and setting the indicated airspeed pointer on the Mach number graduation. Two readings shall be taken, the first before the indicator is vibrated, the second after the indicator has been vibrated and reset to the Mach number graduation. The difference between readings is the friction and shall not be greater than that noted in Table V. The second reading is related to the Mach number scale error and shall be within the limits of Table VII.

4.6.4.2.2 QPL and sampling plan "B" test. The Mach scale shall be tested at each of the points of Table VII. The test shall be conducted, at each specified test point, by establishing the specified altitude pressure and adjusting the airspeed pressure to set the indicated airspeed pointer on the Mach number graduation. Two readings of the applied pitot pressure are to be taken at each test point, the first is recorded when the airspeed pointer is set, the second reading is recorded after the indicator has been vibrated and the airspeed pointer readjusted to the Mach number graduation, during the applied vibration. The two pitot pressure readings shall be converted to airspeed. The difference between readings is the friction error and shall not be greater than that noted in Table V. The second reading shall be compared to the value listed in Table VII. The difference is related to the Mach number scale error and shall be within the limits of Table VII. When desired by the inspector or test activity, the indicator shall be tested for friction and scale error at any point or points of the scale and the errors at each additional points shall be within the limits of Table VII.

4.6.4.3 Pointer zero position error. The zero position of the pointer with no differential pressure applied to the indicator shall be within the values specified in Table V.

4.6.5 Low temperature operation. (This test may be combined with the Low Temperature Exposure Test when sampling Plan "B" or QPL testing is involved.) The indicator shall be properly connected except that no power shall be

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applied. The indicator shall then be subjected to an ambient temperature of -55°C ± 2°C for a period of four hours. During the last 15 minutes of the four hour period, lighting power shall be applied to the indicator. At the end of the four hour period and with the temperature maintained at -55°C ± 2°C the indicator shall be tested for scale errors at every other test point marked with an asterisk in Tables VI and VII beginning with the first asterisk. The error at the test points shall not exceed the tolerances shown in Tables VI and VII. No damage to any part of the indicator shall result from this test.

4.6.6 High temperature operation. (This test may be combined with the High Temperature Exposure Test when Sampling Plan "B" or QPL testing is involved.) The indicator shall be properly connected except that no power shall be applied. The indicator shall then be subjected to an ambient temperature of 71°C ± 2°C for a period of four hours. During the last 15 minutes of the four hour period, lighting power shall be applied to the indicator. At the end of the four hour period and with the temperature maintained at 71°C ±2°C, the indicator shall be tested for scale errors at every other test point marked with an asterisk in Tables VI and VII, beginning with the first asterisk. The error at the test points shall be not greater than the tolerances shown in Tables VI and VII. No damage to any part of the indicator shall result from this test.

4.6.7 Lighting (individual tests). Each indicator submitted for acceptance shall be subjected to the Individual Tests of MIL-L-25467 for red and IPL white and MIL-L-27160 for white lighting as applicable.

4.6.8 Position error. The following tests are to be conducted in their entirety during QPL and Sampling Plan "B" testing. For Individual testing, during Quality Conformance testing, only the points marked by the symbol # need to be checked. The Position Error Test may be combined with the applicable Scale Error Test.

4.6.8.1 Airspeed scale. The indicator shall be held in the normal mounting position and the airspeed reading taken after vibration. Airspeed readings shall also be taken with the indicator held in the following positions:

a. Zero airspeed indicia at the 3 O'clock position.

b. Zero airspeed indicia at the 6 O'clock position.

c. Zero airspeed indicia at the 9 O'clock position.

d. Dial face up and horizontal.

e. Dial face down and horizontal.

The test shall be made at 120 knots, 250 knots, and 500 knots as shown in Table V. The maximum deviation of the readings from the original readings shall not differ by more than the tolerances shown in Table V.

4.6.8.2 Mach number scale. The procedure specified in the Position Error Test (Airspeed Scale) shall be repeated for the Mach number scale except that

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the scale shall be set at a reading of 1.0 Mach. The test shall be made at Mach number 0.9 at sea level, Mach 0.9 at 20,000 feet and Mach 0.9 at 40,000 feet as shown in Table VII. The maximum deviation of the position error readings from the original reading shall be not greater than the tolerances shown in Table V.

4.6.9 Magnetic effect. The indicator shall be properly connected and lighting power applied. The indicator shall be revolved about a short bar-magnet compass with the nearest part of the indicator 5 1/2 inches from the bar-magnet. The compass shall have its compensating magnets removed and shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The indicator shall be revolved in a horizontal plane which is perpendicular to the axis of the bar-magnet. The indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees about its horizontal axis. The deflection of the compass at any of the specified positions shall be not greater than one degree. This test shall be repeated with no power applied.

4.6.10 Damping. Sufficient pressure shall be applied to the pitot connection to produce full scale deflection of the airspeed pointer. The pressure shall suddenly be released. The time required for the airspeed pointer to travel from full scale deflection to the 90 knots graduation shall be greater than 0.8 seconds but less than 2.0 seconds.

4.6.11 Overpressure. The indicator shall be overpressure tested as follows:

a. A pressure of 60 in. Hg. shall be applied to the pitot pressure connection of the indicator for a period of 10 minutes.

b. A pressure of six inches in water shall be applied to the static air connection of the indicator for a period of 10 minutes.

c. Not less than five minutes following the application of the above pressures, the indicator shall be tested for room temperature scale error at the test points marked with an asterisk in Tables VI and VII. The errors at the test points shall not exceed the tolerances for High and Low Temperature shown in Tables VI and VII.

4.6.12 Seasoning. The static air connection of the indicator shall be connected to a suitable source of vacuum and the pitot connection shall remain open to room ambient pressure. The indicator shall be cycled between room ambient pressure and 50,000 feet altitude at a rate not greater than two cycles per minute for 1,000 cycles. Approximately 48 hours following the test, the indicator shall be subjected to the Room Temperature Scale Error Tests. The errors shall be not greater than the High and Low Temperature tolerances shown in Table VI and VII.

4.6.13 Lighting. Each indicator shall be tested to verify that the integral lighting system meets the Sampling Plan "A" requirements of MIL-L-25467 or MIL-L-27160 as appropriate. A White lighted indicator shall

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meet MIL-L-21760 requirements. A Red or IPL white lighted indicator shall meet the requirements of MIL-L-25467 except that for IPL the test voltage shall be 4.5 ± 0.1 volts dc and the color shall be determined as follows: using a spectrophotometer, or other suitable color measuring equipment, the chromaticity coordinates, CIE scale, shall be determined. The coordinates shall be: X not less than 0.48 nor greater than 0.54 and the numerical value of Y-YO not greater than 0.01.

4.6.14 Low temperature exposure. The indicator shall be tested in accordance with MIL-STD-810 Method 502.1, Procedure I. At the conclusion of the 72-hour exposure period and while still at the -54°C (-65°F) temperature, the indicator shall be tested for scale errors as specified in the Scale Error Room Temperature Test except that the test points shall be at every other point in Tables VI and VII indicated by asterisks, beginning with the first asterisk. The errors at the test points shall be not greater than those specified for the routine low temperature scale error tests. After the indicator has returned to room temperature, it shall meet all the requirements specified in the Individual Tests.

4.6.15 High temperature exposure (+71°C). The indicator shall be tested in accordance with MIL-STD-810 Method 501.1, Procedure I except that the exposure period at +71°C ± 2°C shall be 24 hours. After the indicator has returned to room temperature, it shall be subjected to and meet the requirements specified in the Individual Tests.

4.6.16 Vibration endurance test. The indicator shall be tested in accordance with MIL-STD-810 Method 514.2, Procedure I part 2 for category b.1 equipment. Figure 514.2-2, curve AR, shall apply. The test shall be conducted with the indicator at an altitude of 35,000 feet and an indicated airspeed of 350 knots. The initial reading at the values specified shall be determined with pressure increasing. At the end of the vibration period, the indicator shall be subjected to the friction and scale errors at room temperature. The airspeed and/or Mach number indications shall not have changed by more than ± 2 knots from the values recorded just prior to this vibration test.

4.6.16.1 Vibration functional test. The instrument shall be vibrated in accordance with 4.6.16. A five minute dwell (in lieu of the 20 minute dwell specified in MIL-STD-810) in each axis, at the two most severe resonance frequencies shall be made at the applicable vibration level. The test shall be conducted with the indicator at a test altitude of 35,000 feet and an indicated airspeed of 350 knots. No condition of applied vibration shall cause errors involving airspeed pointer or Mach scale oscillation or variation in excess of the tolerance specified in Table V. In addition, the test shall be repeated for vibration error at 120 knots at or near sea level pressure. The pointer indication shall not oscillate more than ± 2 knots.

4.6.17 Acceleration. Acceleration tests shall be performed in accordance with MIL-STD-810, method 513.2, procedure I for structural test and procedure II for operational test.

4.6.17.1 Operational test, procedure II. The airspeed indicator shall be subjected to tests on a centrifuge while oriented in each of the six positions as specified in MIL-STD-810, method 513.2. The indicator shall be installed

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so that it can be electrically energized and pneumatically operated during the test. The indicator shall be subjected to the "g" levels listed in table VIII. The test shall be performed with the indicator pitot and static pressure ports exposed to a constant differential pressure of approximately 300 knots. The change in airspeed scale errors recorded for each position during acceleration shall be not greater than 1.5 knots per "g". The indicator lighting shall be noted to be continuous (without flicker) during acceleration. The inspections listed in 4.4.1 shall be performed upon completion of this acceleration test.

4.6.17.2 Structural test, procedure I. The airspeed indicator shall be subjected to tests on a centrifuge while oriented in each of the six positions as specified in MIL-STD-810, method 513.2. The indicator shall be electrically energized during this test, and a constant differential pressure of approximately 300 knots shall be maintained at the pressure ports. Indicator accuracy need not be recorded during this test. The indicator shall be subjected to the "g" levels listed in table VIII. Upon completion of this acceleration test, the indicator shall be subjected to the inspections listed in 4.4.1.

TABLE VIII. Operational and structural test "g" levels

MIL-STD-810 Method 513.2 Indicator Position	Centrifuge "g" Test Level	
	Operational	Structural
Fore	2	3
Aft	6	9
Up	9	13.5
Down	3	4.5
Lateral, Right Side	4	6
Lateral, Left Side	4	6

4.6.18 Lighting. With the exception of the Life Test, the indicator shall be subjected to the Sampling Plan "B" Test of MIL-L-25467 or MIL-L-27160, whichever is applicable. If the indicator is white lighted it shall be in accordance with MIL-L-27160 and if it is red or IPL white it shall be in accordance with MIL-L-25467 except that the IPL white shall be tested at 4.5 ± 0.1 volts DC and the color shall be determined by using a spectrophotometer, or other suitable color measuring equipment, to determine the chromaticity coordinates on the CIE scale. The coordinates must be: X not less than 0.48 or greater than 0.54 and the numerical value of Y-YO not greater than 0.01.

4.6.19 Humidity test. The indicator shall be tested in accordance with Method 507.1 Procedure I of MIL-STD-810. After the testing procedure, the indicator shall be subjected to and meet the requirements of the Individual Tests.

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4.6.20 Mounting lugs. The indicator case shall be mounted on a suitable testing machine with the face of the case in a horizontal plane so that the mounting lugs receive no additional support. A suitable pin shall be inserted through the hole in the mounting lug and attached to the pull strap on the testing machine. A load of 175 pounds shall be applied for one minute to each lug in a direction toward the front of the case. The lugs shall with-stand the applied load without fracture, and there shall be no damage to any part of the indicator.

4.6.21 Internal examination. The case of the indicator shall be opened and the mechanism of the indicator shall be examined. The lamp circuit of the indicator shall also be inspected as specified in MIL-L-25467 or MIL-L-27160, whichever is applicable. Any deterioration or damage which could in any manner prevent the indicator from meeting functional maintenance, or service requirements during service life shall be cause for rejection.

4.6.22 Shelf life.

4.6.22.1 Aging. The indicator shall be retained at ambient atmospheric conditions for a period of not less than 30 days after completion of other Sampling Plan "B" Tests and then subjected to the Room Temperature Scale Error Test at the test points marked with an asterisk in Tables VI and VII. The scale errors in this test shall not differ from those obtained 48 hours after the Seasoning Tests by more than the values specified in Table V. The indicator shall be subjected to the Lamp Life Test during the entire 30 day aging period.

4.6.22.2 Lamp life. The indicator shall be subjected to the Lamp Life Test specified in MIL-L-25467.

4.6.23 Fungus resistance. The indicator shall be tested in accordance with Method 508.1 Procedure I of MIL-STD-810 except that the exposure period shall be 14 days. At the option of the procuring activity, a certificate of compliance or tests of component parts of the indicator may be accepted in lieu of, or in addition to, tests of the assembled indicator (see 6.2.1).

4.6.24 Salt spray (fog). The indicator shall be tested in accordance with Method 509.1 Procedure I of MIL-STD-810. After the testing period, the indicator shall be subjected to and meet the requirements of the Individual Tests.

4.6.25 Maintainability demonstration. Maintainability demonstration test shall be conducted in accordance with MIL-STD-471. The tests shall be selected in advance and shall be approved by the procuring activity. The instrument shall have a median value of corrective maintenance tasks (Mct) of 6.0 hours when Test Method 1, Test Plan A, of MIL-STD-471 is used. The max Mct shall be no greater than 8.0 hours.

4.7 Reliability assurance tests. Reliability Assurance Tests shall consist of Reliability Qualification Phase Tests and Manufacturing Run-in Tests. Reliability Qualification Phase Tests shall be conducted in accordance with 4.7.1. Manufacturing Run-in Tests shall be conducted in accordance with 4.7.2. Prior to either test, the indicators shall have first passed the individual tests. Accept-reject criteria shall be in accordance with

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MIL-STD-781 and classification of failures shall be in accordance with MIL-STD-2074.

4.7.1 Reliability qualification phase tests. Prior to approval for listing on the Qualified Products List, three indicators shall be subjected to the following test conditions:

a. Temperature cycling. The equipment shall be temperature cycled between -54 and +71 degrees C. The equipment shall be stabilized at -54 degrees C. The equipment shall then be increased until stabilization is reached at the +71 degrees C level as determined by a thermal survey. The equipment shall be maintained at +71 degrees C for a minimum of two hours after stabilization. The temperature shall then be decreased by chamber cooling until the point of maximum thermal inertia stabilizes at -54 degrees C. This cycle shall continue for the duration of the test.

b. Vibration. Vibration shall be applied perpendicular to the roll and pitch axes of the indicator. The value of the applied acceleration shall have a peak value of 2.2 ± 10 percent (sine) at any non-resonant frequency between 20 and 60 hertz as measured at the mounting point of the indicator.

c. Lighting. Fifty-seven minutes on, three minutes off each hour. Input voltage shall be 5 volts +5 -2 percent.

d. Exercise. During each hour of duty cycling, the indicator shall be pneumatically cycled at a rate of aproximately 4 cycles per hour. The pneumatic cycle shall be applied to the static port with the pitot port left open and shall cause the airspeed pointer to cycle to approximately 750 knots and return to ambient within approximately 15 minutes for a total of at least 4 cycles per hour.

e. Accept-reject levels. For the qualification phase, the accept-reject criteria for MIL-STD-781 Test Plan IIC shall apply.

4.7.2 Manufacturing run-in test. Prior to the manufacturing run-in test, the indicator shall be vibrated perpendicular to the roll and pitch axis for a 10 minute period. The frequency of vibration shall be any frequency between 20 and 60 Hz. The amplitude of vibration shall be approximately 0.5 g as measured on the instrument panel. Each indicator manufactured shall be subjected to a Manufacturing Run-in Test of six hours duration. The first two hours of run-in shall be accomplished with the indicator exposed to a temperature of +71 degrees C. The indicator shall then be returned to a room ambient condition at a minimum rate of change of five degrees C per hour. The remaining four hours of the test shall be accomplished under room ambient conditions with the indicator mounted on a special test table. The table shall subject the indicator to roll, pitch, and yaw motion (Scorsby) at a frequency of five to seven oscillations per minute through an angle of 30 degrees. The 5 volt lighting circuit shall be energized continuously during the test. The indicator shall be pneumatically cycled during the six hour

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run-in test. The pneumatic cycle shall be applied to the static port with the pitot port left open and shall cause the airspeed pointer to cycle to approximately 750 knots and return to ambient within approximately 15 minutes for a total of at least 4 cycles per hour. Failures noted and corrective action taken during the run-in test shall be summarized for the monthly production inspection report (see 6.2.2). Repair or recalibration of the indicator shall require resubmission of the indicator to an additional six hour run-in test until such time as the test is failure free.

4.7.3 Preventive maintenance. Once an indicator has been submitted for Reliability Assurance Tests no preventive maintenance shall be performed on the indicator prior to completion of that particular phase of Reliability Assurance Tests, Qualification or Sampling, for which it was intended, has been completed.

4.7.4 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, shall be part of the test procedures to be made available by the contractor for approval prior to the beginning of the Reliability Assurance Tests.

4.7.4.1 Approval. For Qualified Products Listing the test details shall be submitted to the qualifying activity for approval. When submitted in accordance with contractual requirements approval is made through the procuring activity.

4.7.5 Equipment failure. Should a failure occur during either the Sampling, Reliability Assurance or Special Tests, the following action should be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

c. Implement corrective action to preclude recurrence of the failure.

4.8 Test procedures. The procedures used for conducting Quality Conformance Tests shall be prepared by the contractor and made available to the procuring activity for review and approval (see 6.2.2). The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract (see 6.2.1). MIL-T-18303 shall be used as a guide for Preparation of test procedures. When approved test procedures are available from previous contracts such procedures may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures including additional tests when deemed necessary (see 6.2.1).

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5. PACKAGING.

5.1 Packaging.

5.1.1 Level A. The indicator shall be packaged in accordance with MIL-STD-794. The method of preservation shall be in accordance with MIL-P-116, Method IId using metal reusable containers.

5.1.2 Level C. When this level is required, packaging shall be in accordance with MIL-STD-1188.

5.2 Packing.

5.2.1 Levels A and B. The indicator shall be packed in accordance with MIL-STD-794. The level or levels required shall be as specified in the invitation for bid or contract.

5.2.2 Level C. The indicator shall be packed to insure that the shipment arrives in a satisfactory condition at destination. The shipment shall conform to the applicable carrier rules and regulations in effect at the time of the shipment.

5.3 Marking. The interior and exterior containers shall be marked as specified in MIL-STD-794.

5.3.1 Precautionary marking. The following precautionary markings shall appear conspicuously on two sides of each interior package and shipping container where practicable.

FRAGILE

DELICATE INSTRUMENTS

HANDLE WITH CARE

5.4 Face plate protector. A Face Plate Protector, NSN 1RM-6610-00-138-8048-UX, must be supplied with each MS17264 indicator. No indicator shall be accepted by the Government unless the proper Face Plate Protector is supplied with the indicator and properly attached. It is intended that the Face Plate Protector remain with the indicator until the indicator is installed in an aircraft, except for temporary removal for inspection and testing when necessary (see 6.2.1).

6. NOTES

6.1 Intended use - The indicator covered by this specification is intended for use on high performance aircraft to give an indication of airspeed and mach number by means of a single indicator.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number, and date of this specification.

b. The quantity and MS part number which specifies the type of lighting, presence or absence of mach number setting index and static tubing size.

c. Levels of packaging (see 5.1) and packing (see 5.2) desired.

d. Panel lighting color to be used (see 3.5).

e. That the Face Plate Protector is available through the Naval Aviation Supply Office and method by which it may be obtained (see 5.4).

f. Designate if contractor is to furnish durable black finish round head brass machine screws and removeable spring nuts for indicator installation. The screw length should also be defined (see 3.10.1).

g. Designate Naval test facility where test samples shall be submitted (see 4.4).

h. Whether fungi tests of component parts or assembled indicator is to be accomplished (see 4.6.24).

i. The frequency of Reliability Test Reports.

j. If lot size for reliability sampling phase is not as specified in 4.7.2.1.

k. The preparation of test procedures for review and approval in accordance with 4.8. Purchasers of this equipment should consult with the cognizant NAVAIRSYSCOM representative on matters pertaining to the review and approval of such test procedures.

l. For any aircraft other than the F-14 an adapter to reduce the static fitting from 3/8 to 1/4 inch.

m. If test procedures are to be modified or expanded (see 4.8).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporate a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

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Paragraph no.	Data requirement title	Applicable DID no.
4.4, 4.7.2	Reports, Production Inspection	DI-T-4904
4.4.3,4.8	Procedures, Test	UDI-T-21347

(Data item descritions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be elgible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Commanding Officer, Naval Avionics Center, Code 932, 6000 East 21st St., Indianapolis, IN 46218, and information pertaining to qualification of products may be obtained from that activity.

6.4 Definitions.

6.4.1 Scale error. By the "scale error" is meant the differences between the indication of the instrument and the calibrated airspeed corresponding to the applied differential pressure as specified in MIL-STD-1524.

6.5 Drawing requirements. The attention of the contractor is invited to the fact that Government contracts for equipment, whether acquired directly by the Government or through another contractor, require that all drawings submitted must be in accordance with MIL-D-1000.

Preparing Activity
Navy-AS
Project No. 6610-N056

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-I-23671B(AS)	Indicator, MS17264, Airspeed and Mach Number, Integrally Lighted, Pitot-Static 80 - 850 Knots

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

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